Lecture 6: Time Series Estimation Techniques

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Overview

- So far: population arguments for **identification** of SVMA coefficients
  - Basic idea: use addt'l identifying information (e.g., invertibility + X, IVs, ...) to assign a causal interpretation to certain estimable moments of aggregate time series data
  - Open **Q**: what econometric techniques should we use to estimate those moments?
- This lecture: **VARs** vs. **local projections**
  - Will show that they are nothing more than two different techniques for estimating second moments. In population, under the same structural ID asn's, they yield the same objects
  - Finite-sample recommendations: should we use LPs? VARs? something else?

Outline

1. VARs vs. Local Projections
   - Example I: recursive identification
   - Example II: IV identification

2. Practical Guidance
   - Bias-Variance Trade-Off
   - Resolving The Trade-Off

3. Summary

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VARs vs. Local Projections

- Dominant methods in semi-structural time series: **VARs** and **Local Projections**
  Going back to Sims (1980) and Jordà (2005).
  - Can find a lot of claims in the prior literature that these are somehow **fundamentally different methods**. We will show that this is not the case.
- The argument will proceed in two steps:
  1. Define **VAR** and **LP** estimators
  2. Show that their estimands are the same, using **linear projection arguments**

Note: none of this will rely on an underlying SVMA model. The point of the SVMA is instead just to establish that this common LP/VAR estimand is actually structurally interesting.

Setting and estimators

- **Data-generating process**
  - Sole as'n: data $\{w_t\}$ are covariance stationary and strictly nondeterministic, with absolutely summable Wold representation coefficients and an everywhere-nonsingular spectral density
  - Split the data as $w_t = (r_t', x_t, y_t, q_t')'$ where r_t and q_t are "controls", and we are interested in the response of y_t to an "impulse" to x_t
- **Impulse response estimators**
  1. **LP IRF estimator**
     - A local projection is a regression of an outcome y_t on a "shock" x_t plus controls
     - The LP(∞) regression equation for horizon $h = 0, 1, 2, \ldots$ is

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' r_t + \sum_{\ell=1}^{\infty} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t}$$

     - The LP IRFs are the $\{\beta_h\}$

Setting and estimators

- **Impulse response estimators**
  2. **VAR IRF estimator**
     - Consider the reduced-form VAR(∞)

$$w_t = c + \sum_{\ell=1}^{\infty} A_\ell w_{t-\ell} + u_t$$

     - Under a recursive ordering, we arrive at the following "structural" VAR:

$$A(L)w_t = c + B\eta_t$$

$B = \text{chol}(\text{Var}(u_t))$, $\eta_t \equiv B^{-1}u_t \sim WN(0, I)$, $\eta_{1,t} \propto u_{1,t}$, $\eta_{2,t} \propto u_{2,t} - \mathbb{E}^*(u_{2,t} \mid u_{1,t})$, ...

     - Let $C(L) = A(L)^{-1}$. We can then write

$$w_t = \chi + C(L)u_t = \chi + \sum_{\ell=0}^{\infty} C_\ell B\eta_{t-\ell}, \quad \chi \equiv C(1)c$$

Then the VAR IRF of y_{t+h} to an innovation to x_t is

$$\theta_h \equiv C_{n_r+2,\bullet,h} B_{\bullet,n_r+1}$$

Equivalence result

Proposition

Let $\tilde{x}_t \equiv x_t - \mathbb{E}^*\left[x_t \mid r_t, \{w_\tau\}_{-\infty<\tau<t}\right]$. *Then*

$$\theta_h = \sqrt{\mathbb{E}(\tilde{x}_t^2)} \times \beta_h, \quad h = 0, 1, \ldots \tag{1}$$

- (1) states that **LPs** and **VARs** estimate the same impulse responses
  - Note that so far we haven't assumed an underlying SVMA model. That will only be needed to argue that the object in (1) is structurally interesting.
  - The factor of proportionality in (1) just reflects a different scaling normalization, with VARs normalizing impulses to have unit variance. Will see that in the proof.
- Proof intuition: both techniques just estimate **certain linear projections** (= functions of 2nd moments) and are willing to give them a "structural" interpretation

Equivalence result: proof sketch

- By the Frisch-Waugh-Lovell theorem, the **LP estimand** is

$$\beta_h = \frac{\mathrm{Cov}(y_{t+h}, \tilde{x}_t)}{\mathbb{E}(\tilde{x}_t^2)}$$

- The **VAR impulse responses** equal

$$\theta_h = C_{n_r+2,\bullet,h} B_{\bullet,n_r+1} = \mathrm{Cov}(y_{t+h}, \eta_{x,t})$$

where $\eta_t = (\eta'_{r,t}, \eta_{x,t}, \eta_{y,t}, \eta'_{q,t})'$. By the properties of Cholesky decompositions:

$$\eta_{x,t} = \frac{1}{\sqrt{\mathbb{E}(\tilde{u}_{x,t}^2)}} \times \tilde{u}_{x,t}$$

where $u_t = (u'_{r,t}, u_{x,t}, u_{y,t}, u'_{q,t})'$ and $\tilde{u}_{x,t} \equiv u_{x,t} - \mathbb{E}^*(u_{x,t} \mid u_{r,t}) = \tilde{x}_t$

- Comparing the above relations, the result follows

Equivalence result: discussion

- Plagborg-Møller and Wolf (2021) further discuss the scope of this equivalence result
  1. Previous argument was for recursive SVAR(∞) estimands. Can easily show that the same logic works for **non-recursive identification**: VAR shock $= b'u_t$ so we could run the LP

$$y_{t+h} = \mu_h + \beta_h(b'w_t) + \sum_{\ell=1}^{\infty} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t}$$

  2. Can also show: with p lags (rather than ∞), equivalence $\approx$ **up to horizon** p

- The generality of the result reflects its very **simple intuition**:
  - VAR(p) IRF = mean-square optimal forecast given the second moments implied by the VAR(p) model. But a VAR(∞) matches all second-moment properties of the data.
  - Thus VAR IRF = optimal forecast given second moments of the data = LP

- Next: walk through **LP**/**VAR** implementations of some canonical identification schemes

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VAR vs. LP: recursive identification

- Recall the identification scheme of Christiano-Eichenbaum-Evans (1999)
  - Assume **SVMA model** + **invertibility** + **recursive ordering** of macro var's, consistent with slow-moving real effects of monetary policy
- How could we implement this as a **local projection**?
  - Note that the model fits immediately into our structure from before: r_t = GDP, prices, ..., x_t = federal funds rate, q_t = profits, M2, y_t = any variable in VAR
  - We could thus consider running the **LP**

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' r_t + \sum_{\ell=1}^{p} \delta_{h,\ell}' w_{t-\ell} + \xi_{h,t}$$

- Let's see the equivalence result in action by computing the common "recursive ordering" **LP**/**VAR** estimand in a simple example DGP [details not important here]

VAR vs. LP: recursive identification

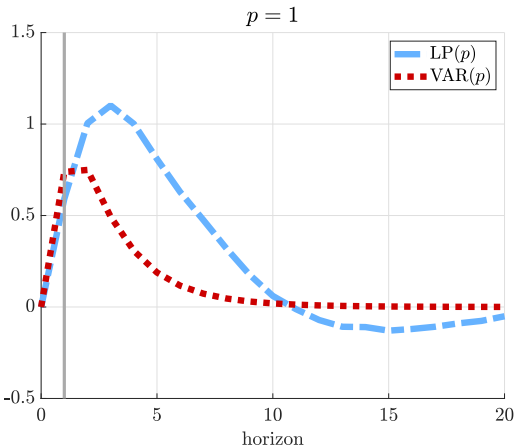
$p < \infty$: same shock so same IRFs at $h = 0$, but then VAR extrapolates

Can show: VAR $\approx$ LP for impulse response horizons $h \leq p$.

VAR vs. LP: recursive identification

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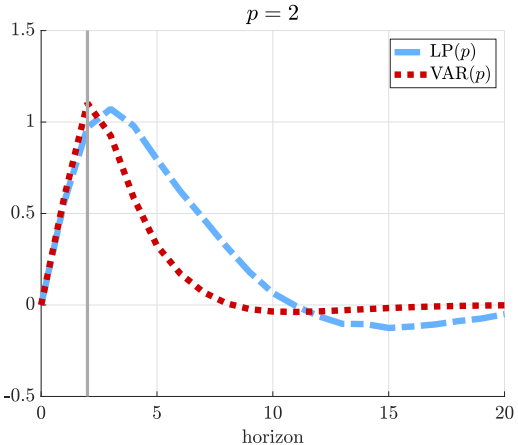
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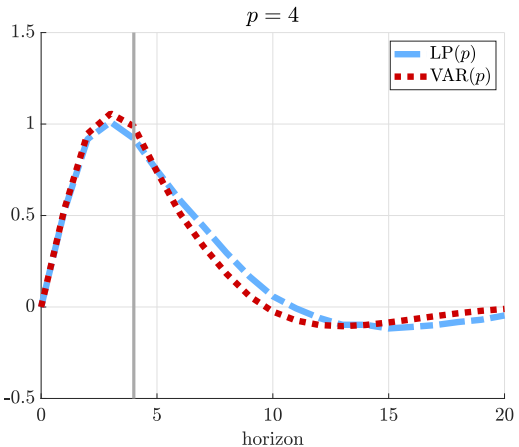
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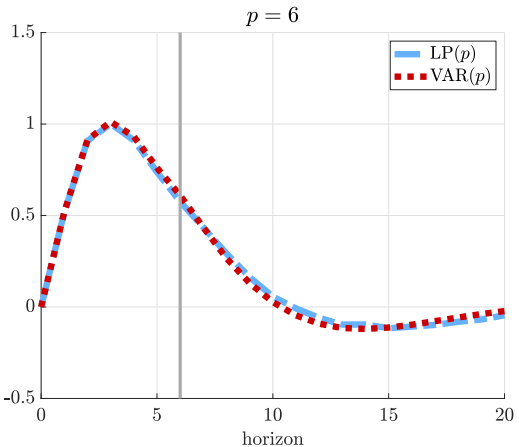
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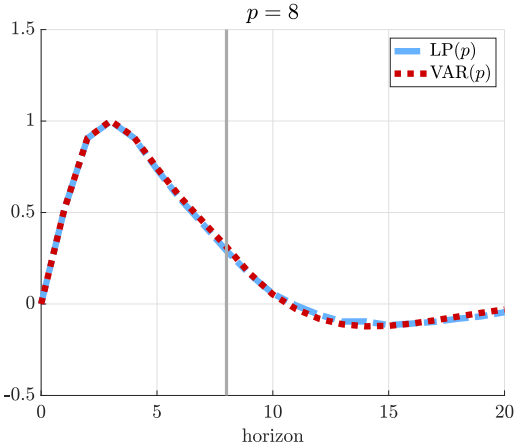
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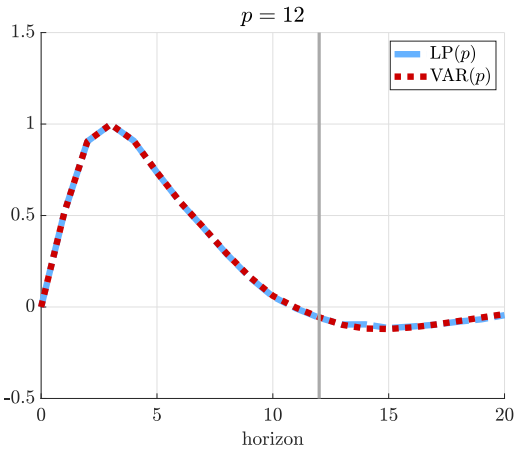
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VAR vs. LP: IV identification

- Now return again to the **SVMA-IV model**

$$\begin{aligned} w_t &= \Theta(L)\varepsilon_t \\ z_t &= \alpha\varepsilon_{1,t} + \sigma_\nu \nu_t, \quad (\varepsilon_t', \nu_t)' \sim WN(0, I) \end{aligned}$$

- We have seen that relative IRFs are identified. How could we estimate them?
- Here the typical approach is an **LP** implementation, known as **LP-IV**:
  - Basic idea: use z_t to instrument for x_t in a projection of y_{t+h} on x_t

$$y_{t+h} = \mu_h + \beta_h x_t + \text{controls} + \xi_{h,t}$$

  - Will decompose this into reduced-form and first-stage to arrive at equivalent VAR representation, using our previous results

VAR vs. LP: IV identification

- **LP-IV**
  - Let $W_t = (z_t, w_t')'$. The reduced-form and first-stage projections are

$$y_{t+h} = \mu_{RF,h} + \beta_{RF,h} z_t + \sum_{\ell=1}^{\infty} \delta_{RF,h,\ell}' W_{t-\ell} + \xi_{RF,h,t}$$

$$x_t = \mu_{FS} + \beta_{FS} z_t + \sum_{\ell=1}^{\infty} \delta_{FS,\ell}' W_{t-\ell} + \xi_{FS,t}$$

  - We thus get relative **LP-IV** IRFs as $\beta_h \equiv \beta_{RF,h}/\beta_{FS}$
- **Recursive VAR**
  - From previous results: can recover $\beta_{RF,h}$ and β_{FS} from a recursive VAR in $W_t = (z_t, w_t')'$
    Note: called "internal-IV VAR," vs. "proxy/external-IV VAR" last time.
  - Aside: this is a VAR that works without **invertibility**. Why? non-invertibility is related to measurement error $\sigma_\nu \nu_t$, which merely induces a constant attenuation bias

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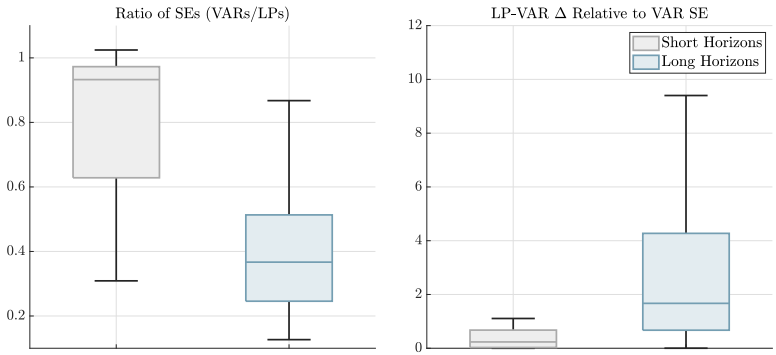
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LPs vs. VARs in finite samples

How do **VARs** and **LPs** differ in finite samples?

Figure is based on replication of empirical applications in Ramey (2016).



suggestive of a *bias-variance trade-off* ...

LPs vs. VARs in finite samples

- Theory so far: **LPs** = **long-lag VARs**
- In practice: what if we use **small-to-moderate lag lengths**?
  - Simple intuition: short-lag VARs imply **addt'l shrinkage**. Extrapolate long-run relations from short-run autocovariances.
  - This suggests a **bias-variance trade-off**, consistent with the previous figure. Now:
    1. Very quick sketch of analytics of the bias-variance trade-off. "No free lunch" for VARs.
    2. Resolving the trade-off in practice: MSE vs. coverage vs. decision theory

  Main references: Montiel-Olea, Plagborg-Møller, Qian, and Wolf (2026a,b).

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Setting

- Data-generating process: **VAR(p_0)** + **MA(∞)**
  - Observables w_t satisfy

$$w_t = \sum_{\ell=1}^{p_0} A_\ell w_{t-\ell} + H \left[I + \sum_{\ell=1}^{\infty} \alpha_\ell L^\ell \right] \varepsilon_t$$

    Restrict **MA(∞)** component to be "small" vs. sample size T = hard to detect statistically.
  - Assume straightforward identification, e.g., H is lower-triangular, and denote true horizon-h IRF by θ_h. Challenge: residuals of VAR with p_0 lags aren't equal to the true shocks ε_t.
- Now compare **two estimators**:
  1. **VAR** with $p \geq p_0$ lags and correct identification scheme. Denote IRF by $\hat{\theta}_h^{\text{VAR}}$.
  2. **LP** with $p \geq p_0$ lags and correct identification scheme. Denote IRF by $\hat{\theta}_h^{\text{LP}}$.

For full formal description see Montiel-Olea, Plagborg-Møller, Qian, and Wolf (2026a).

Bias-variance trade-off

Proposition

*The **LP** and **VAR** estimators have the following distributions:*

$$\hat{\theta}_h^{VAR} \dot{\sim} N\left(\theta_h + b_h(p), \tau^2_{h,VAR}(p)\right), \quad \hat{\theta}_h^{LP} \dot{\sim} N\left(\theta_h, \tau^2_{h,LP}\right), \tag{2}$$

where $\dot{\sim}$ denotes "approximate distribution in large samples".

1. **Robustness.** *The **LP** estimator is approximately unbiased, while the **VAR** estimator is generally asymptotically biased ($b_h(p) \neq 0$).*
2. **Precision.** *The **VAR** variance is weakly below that of **LP** ($\tau^2_{h,VAR}(p) \leq \tau^2_{h,LP}$).*
3. **Equivalence.** *Bias and variance of **VAR** and **LP** are identical for $h \leq p - p_0$.*
   Thus: VAR bias = 0 if we control for sufficiently many lags, but then also has to be as noisy as LP.

How bad can things get in theory?

Proposition

Let T denote the sample size and $\mathcal{M}$ denote the fraction of the variance of the moving average residual that is explained by the lagged shocks. Then

$$\frac{|b_h(p)|}{\tau_{h,VAR}(p)} \leq \sqrt{T \times \mathcal{M}} \times \sqrt{\frac{\tau^2_{h,LP}}{\tau^2_{h,VAR}(p)} - 1}, \tag{3}$$

and this bound is attained for some $\alpha(L)$.

E.g.: if $T = 100$, $\mathcal{M} = 0.01$, $\tau_{h,\text{VAR}}(p)/\tau_{h,\text{LP}} = 0.5$, then the bias can be 1.73 as big as the standard error.

- Interpretation: there is "no free lunch" for **VARs**
  1. The *only* way to ensure robustness is equivalence with **LP** [i.e., via many lags]
  2. If the **VAR** is much more precise, then it is *necessarily* vulnerable to misspecification

How does the trade-off look in practice?

- **Strategy**: empirically calibrated, large-scale simulation study Details

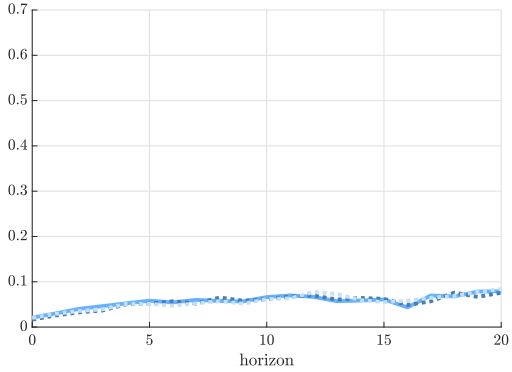
  Builds on and extends our earlier simulation work in Li, Plagborg-Møller, and Wolf (2024).

  - Use Stock-Watson dynamic factor model, ensuring good match to U.S. macro time series.
    Consider both stationary and non-stationary variants. Include ARCH errors for realism.
  - Construct lots of DGPs: randomly draw subsets of observables from the DFM series, then implement observed shock or recursive ID schemes for monetary & fiscal shocks
  - Simulate data, then estimate **LPs**, **VARs**, and several of their variants, and compare

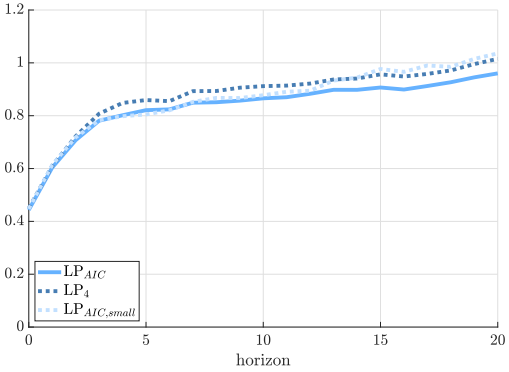
    to be useful for applied work, an **estimation procedure** should *at least* work well here

- **Next slide**: estimation results averaging across stationary & non-stationary DGPs

Simulation evidence

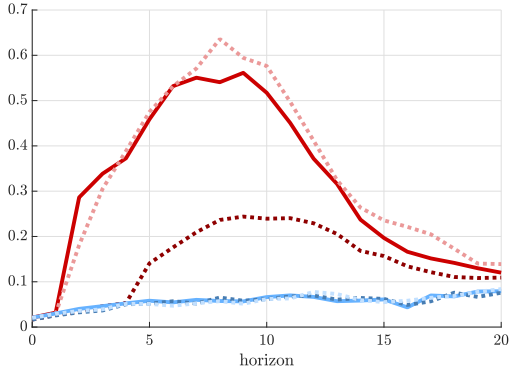


Bias

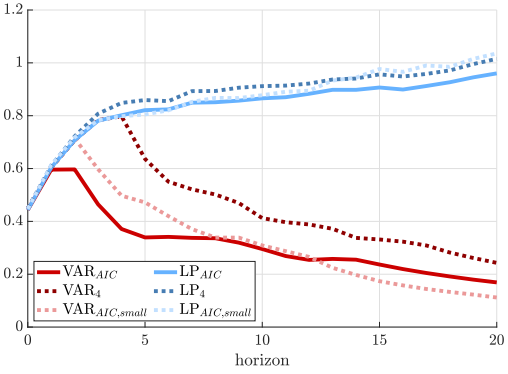


Standard deviation

Simulation evidence

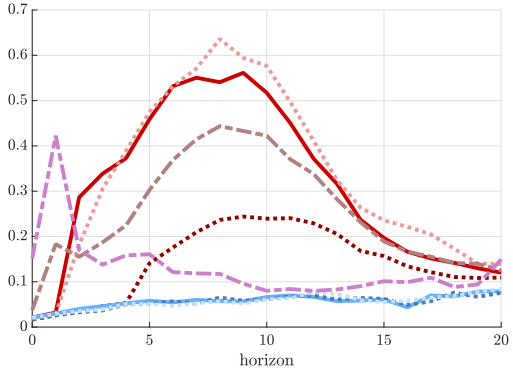


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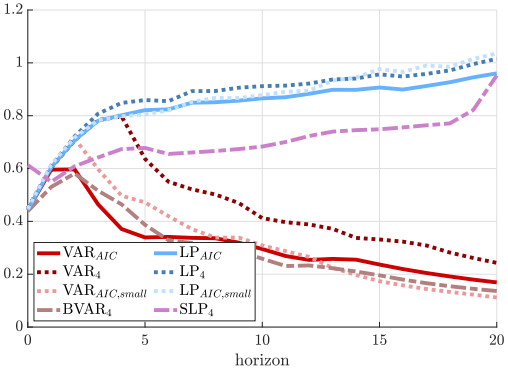


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Simulation evidence



Bias



Standard deviation

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What to do?

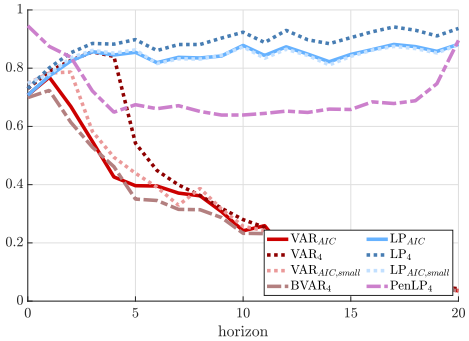
- Clearly need an **objective function** to judge the trade-off
- We will consider three such criteria:
  1. **Mean-squared error**
     - Defined as
     $$\text{MSE}(\hat{\theta}_h) = \underbrace{\left[\mathbb{E}\left(\hat{\theta}_h - \theta_h\right)\right]^2}_{\text{Bias}(\hat{\theta})^2} + \text{Var}(\hat{\theta}_h)$$
     - Natural criterion for *point estimation*—which estimator is most accurate on average?
  2. **Coverage**
     - Def'n: how likely is an x% confidence interval to indeed contain the truth?
     - Natural criterion for *inference*—which estimator yields CIs that are robustly valid?
  3. **Decision theory**: downstream use of the estimated $\hat{\theta}$'s for optimal policy design

MSE

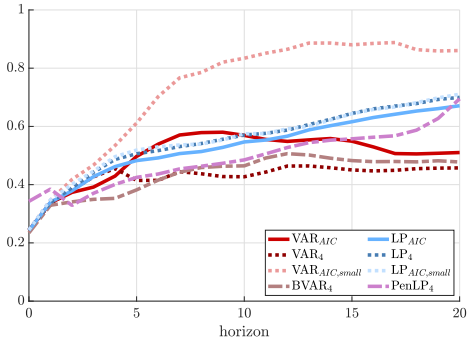
- Theory for **worst case**: $\mathbf{MSE_{VAR}} = \mathbf{MSE_{LP}}$ already for $T \times \mathcal{M} = 1$

  And this is a very small, hard-to-detect amount of misspecification.

- But in practice: typical DGP not that close to worst case, so on average **VAR** is better

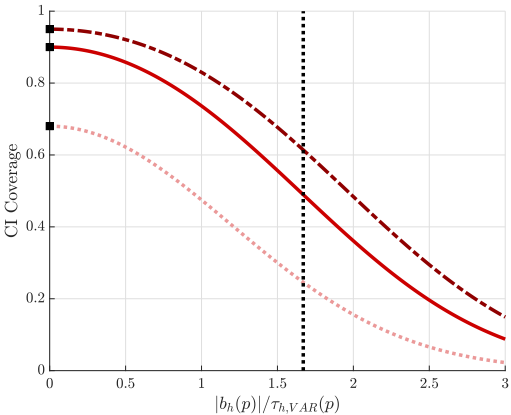


MSE for Stationary DGPs



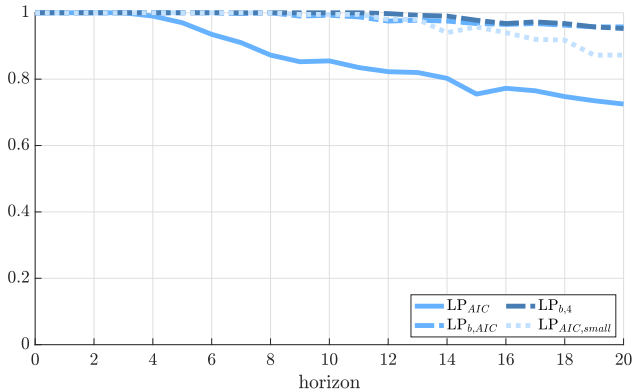
MSE for Non-stationary DGPs

Bias and coverage



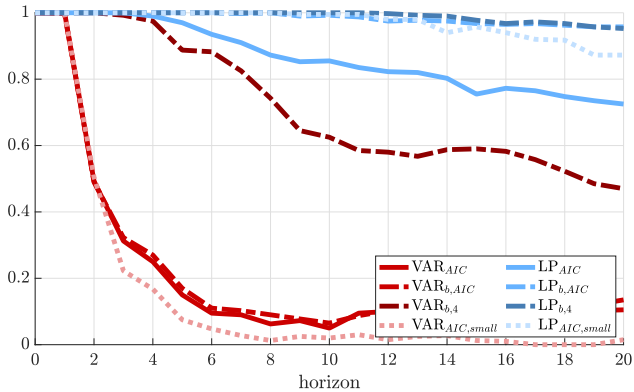
- Natural to require correct **coverage**
  - Given small samples in macro, no empirical study is complete w/o uncertainty assessment
  - Want CIs that are *robustly* valid—not just accurate for the av'g DGP, but across DGPs
- Challenge for **VARs**: bias is really costly for coverage! CI has correct width, but off-center.
- Remember: worst-case bias under natural asn's $= 1.73$ as big as standard error

Simulation evidence



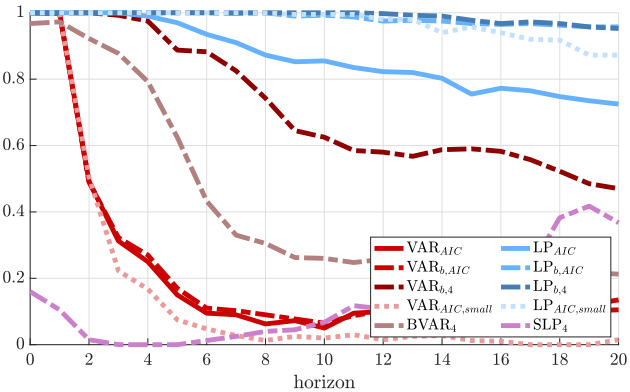
Fraction of DGPs with close-to-correct coverage

Simulation evidence



Fraction of DGPs with close-to-correct coverage

Simulation evidence



Fraction of DGPs with close-to-correct coverage

Decision theory

- **MSE** & correct **coverage** are natural general-purpose criteria in econometrics
- But what's the actual **downstream use** of shock causal effects?
  - Remember Lecture Note 4: policy causal effects Θ_ν as "sufficient statistics" for how policy decisions affect the economy. So natural input for optimal policy design!
    Recent application: Barnichon-Mesters (2023). Strategy: just plug in point estimates $\hat{\Theta}_\nu$.
  - How should a policymaker use empirical estimates $\hat{\theta}_h^{\text{VAR}}$ & $\hat{\theta}_h^{\text{LP}}$ for policy shocks?
    Ongoing work: Montiel-Olea, Plagborg-Møller, and Wolf (2026). Only sketch key takeaways here.
    - → Setting: optimal policy design problem given noisy estimates of policy causal effects
    - → Result I: bad idea to act certainty-equivalent and just plug in the point estimate $\hat{\Theta}_\nu$. If uncertainty is large, you should act cautiously.
    - → Result II: bias hurts more than variance, so prefer to use **LP**

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Summary & takeaways

- This lecture: **estimation methods** for macroeconomic shocks
  - Conceptually: **LP**-**VAR** equivalence [with many lags]
  - In practice: **short-lag VARs** achieve shrinkage in a way that leads to vulnerabilities
    Bias-variance trade-off. Concern for coverage, or policy problem = more concerned about bias.

takeaway: run LPs/longer-lag VARs

- Next: brief discussion of **non-linear** time series methods

Appendix

Encompassing model

- **Dynamic Factor Model** (DFM) [follows Stock-Watson (2016)]

$$
\begin{aligned}
f_t &= \Phi(L) f_{t-1} + H\varepsilon_t \\
X_t &= \Lambda f_t + v_t \\
v_{i,t} &= \Gamma_i(L) v_{i,t-1} + \Xi_i \xi_{i,t}
\end{aligned}
$$

  Gaussian aggregate shocks ε_t + idiosyncratic noise $\xi_{i,t}$.

  - f_t: six latent driving factors, evolve as VECM or VAR, driven by six aggregate shocks ε_t.
  - X_t: 207 quarterly macro time series, spanning various categories.
  - $v_{i,t}$: idiosyncratic noise, evolves as AR(4), independent across i.

- Parameters fixed at estimates from U.S. data in Li et al. (2024)
- Estimate ARCH processes for the errors $\{\varepsilon_t, \xi_{i,t}\}$

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Lower-dimensional DGPs and estimands

- Draw subsets of 5 variables $\bar{w}_t \subset X_t$. DFM implies that $\bar{w}_t$ follows VAR(∞).
  - For each $\bar{w}_t$ VAR(∞) we will define monetary & fiscal policy structural shock estimands
  - $\bar{w}_t$ always contains at least one real activity and one price series, + policy instrument i_t (either fed funds rate [monetary shocks] or gov't spending [fiscal shocks])
  - Restrict attention to "salient" series. Select response variable $y_t \in \bar{w}_t$ at random (not i_t).
- Estimands for two structural identification schemes:
  1. **Observed shock** $\varepsilon_{1,t}$. Impulse response estimand: $\theta_h = \frac{\partial y_{t+h}}{\partial \varepsilon_{1,t}}$, $h = 0, 1, 2, \ldots, 20$.
     $H = \frac{\partial f_t}{\partial \varepsilon_t'}$ chosen to maximize impact response of i_t wrt. $\varepsilon_{1,t}$.
  2. **Recursive**. Recursive ordering of $\bar{w}_t$, fiscal shock ordered first, monetary shock ordered last.

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